
MODEL SELECTION AND VALIDATION REPORT (remove_8_biomass model)

OBSERVATIONS:1687

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_8_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.21	0.17
remove_8_tric	0.55	0.03
remove_8_teve	-0.41	0.03
sbt_mean	0.39	0.09
sbt_sd	0.04	0.05
sst_sd	0.04	0.04
log_chla_mean	0.06	0.04
depth	0.20	0.05
log_sum_fishing	0.06	0.03

Dispersion parameter: 0.95

Spatiotemporal AR1 correlation (rho): 0.92

Matérn range: 204.04

Spatiotemporal marginal AR1 SD: 0.84

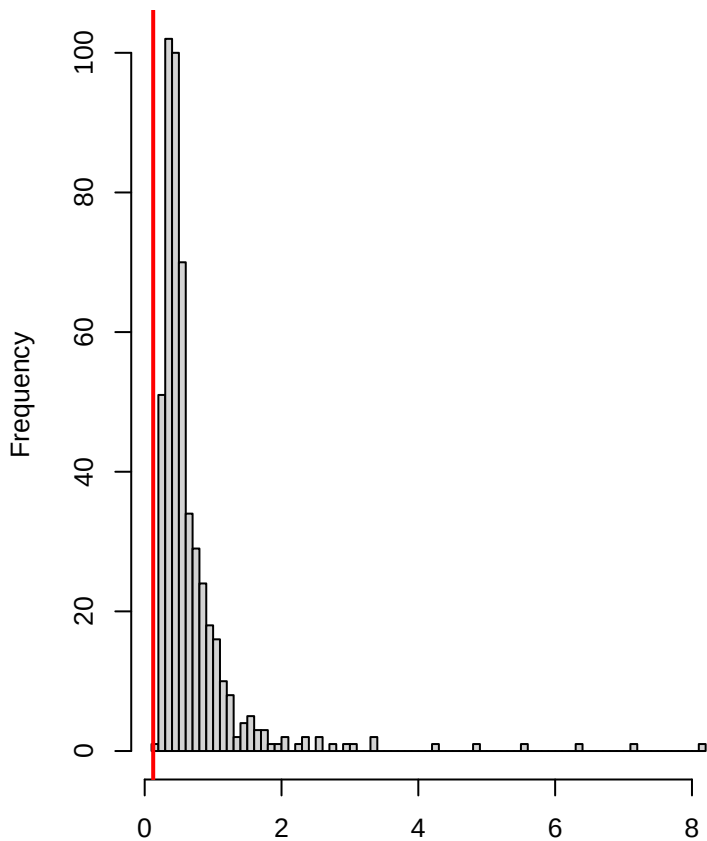
REML criterion at convergence: 10417.465

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

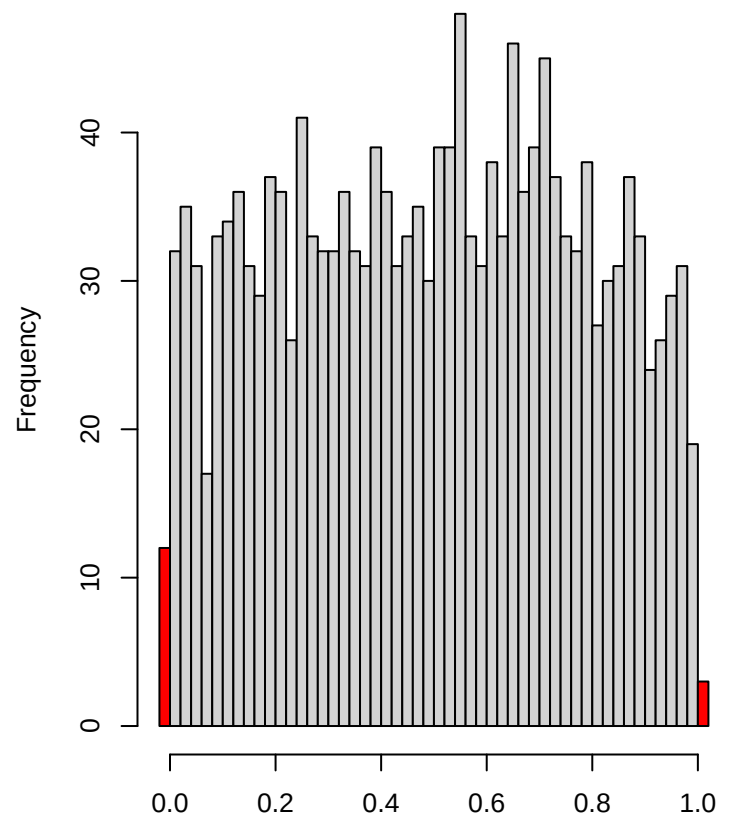
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	20860.93	2.105023
spatiotemporal_random_fields_ar1_with_anisotropy	15	20863.04	75.689460
spatial_random_fields	12	20938.73	3.397726
spatial_random_fields_with_anisotropy	14	20942.12	95.651115
spatiotemporal_random_fields_iid_with_anisotropy	14	21037.77	2.673521
spatiotemporal_random_fields_iid	12	21040.45	145.083465
error_only	10	21185.53	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



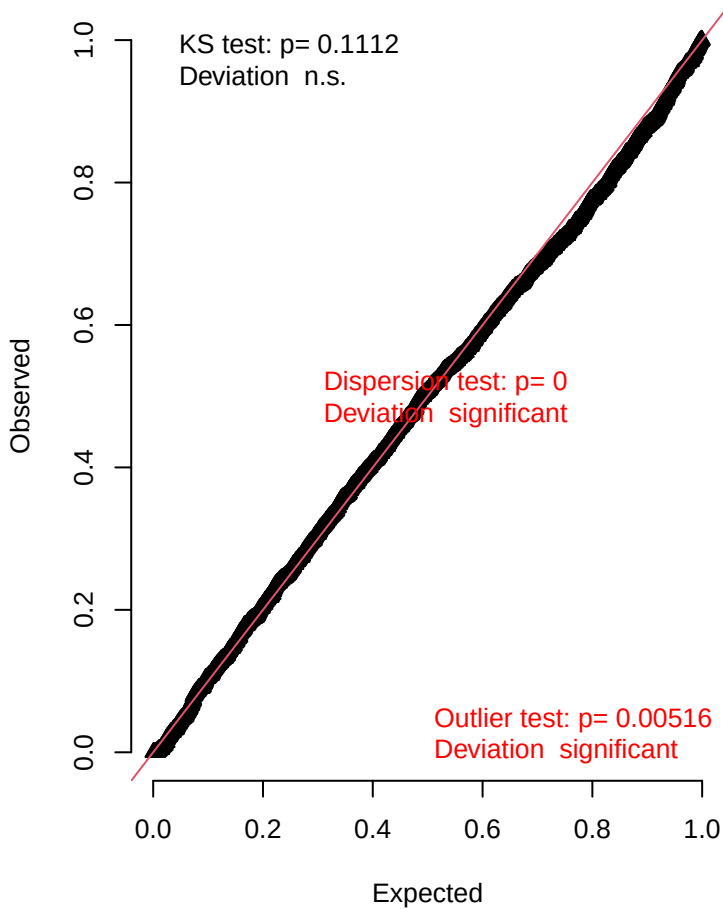
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

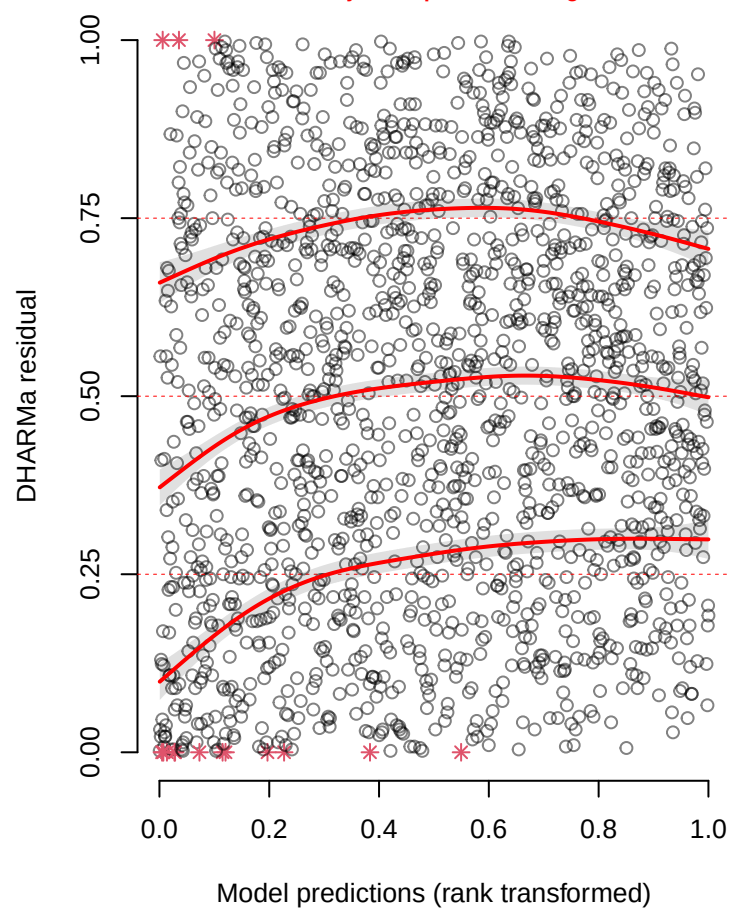


Residuals (outliers are marked red)

QQ plot residuals

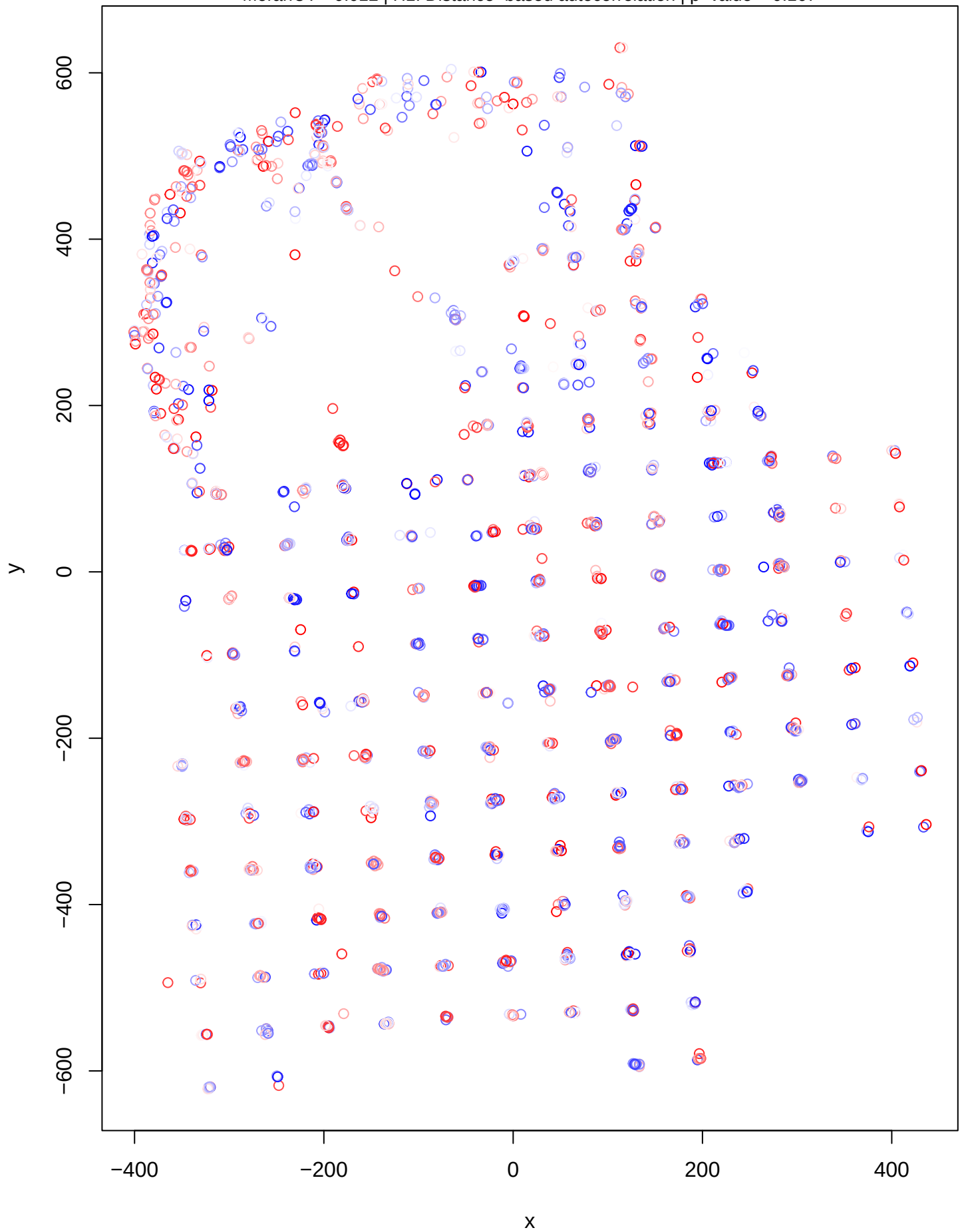


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant

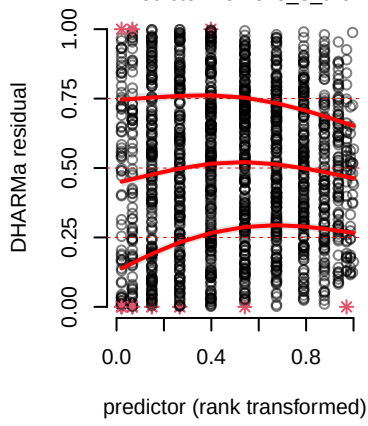


DHARMA Moran's I test for distance-based autocorrelation

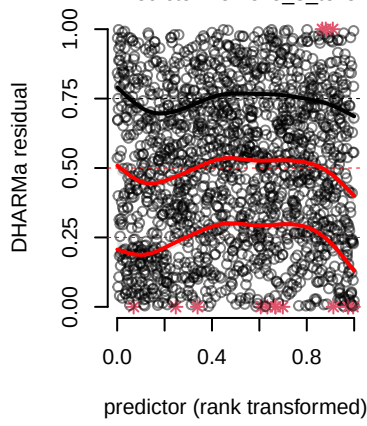
Moran's I = 0.012 | H1: Distance-based autocorrelation | p-value = 0.167



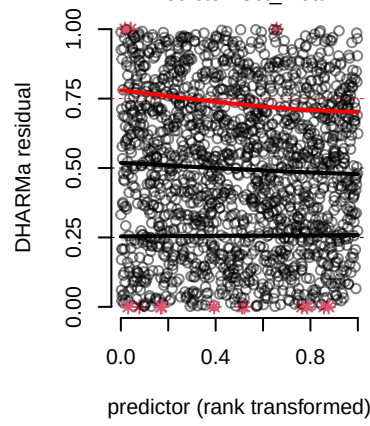
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



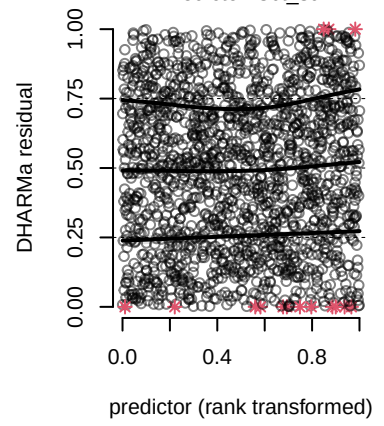
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



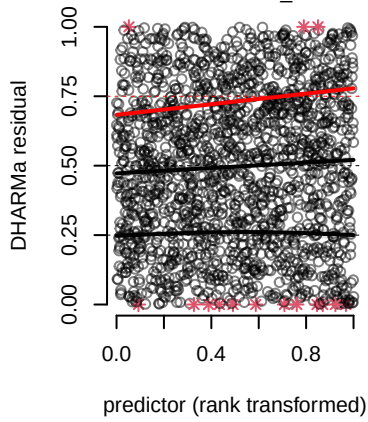
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.



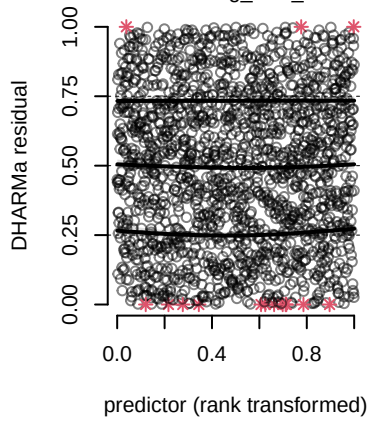
DHARMA residual Residual vs. predictor
No significant problems detected



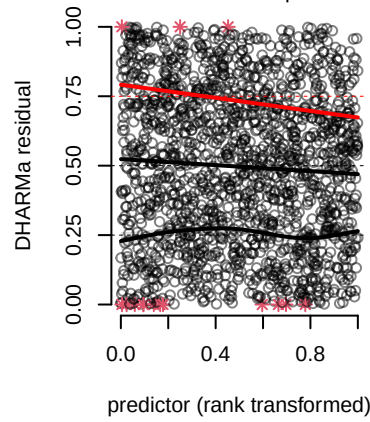
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA residual Residual vs. predictor
No significant problems detected



DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA residual Residual vs. predictor
No significant problems detected

